Introduction

When multiple estimators are available for the same parameter, we need criteria to choose among them. Three classical criteria – unbiasedness, consistency, and efficiency – cover the most important properties: whether the estimator is correct on average (unbiasedness), whether it approaches the truth as more data arrive (consistency), and whether it uses the data as effectively as possible (efficiency).

Prerequisites

The reader should be comfortable with the notions of expectation, variance, and convergence in probability.

Theory

Unbiasedness. An estimator $\hat{\theta}$ is unbiased for θ if $E[\hat{\theta}] = \theta$ for every value of θ in the parameter space. Examples: the sample mean is unbiased for the population mean; the sample variance with divisor $n-1$ is unbiased for σ^2; the plug-in variance with divisor n is biased.

Consistency. An estimator $\hat{\theta}_n$ is consistent for θ if it converges to θ in probability as $n \to \infty$:

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Unbiasedness and consistency are distinct: the sample median is consistent for the population median but can be biased in small samples; the MLE of σ^2 (divisor n) is biased in every finite sample but consistent.

Efficiency. Among unbiased estimators, the **efficient** one is the one with the smallest variance. The Cramer-Rao lower bound (CRLB) sets a floor on this variance:

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)},$$

where $I(\theta)$ is the Fisher information for θ. An estimator that attains this bound is **UMVUE** (uniformly minimum-variance unbiased estimator). For regular models, the MLE is asymptotically efficient: its variance approaches the CRLB as n grows.

These three properties can be in tension. A biased estimator with tiny variance can have lower MSE than an unbiased estimator with large variance; a consistent estimator may converge slowly or may not exist in closed form.

Assumptions

These properties are defined within a model. Change the model, and an estimator that was unbiased, consistent, and efficient for one parameter may be none of the three for another.

R Implementation

```r
library(ggplot2)
set.seed(2026)

simulate_props <- function(n, reps = 5000) {
  mu <- 10
  sigma <- 3
  m  <- replicate(reps, mean(rnorm(n, mu, sigma)))
  md <- replicate(reps, median(rnorm(n, mu, sigma)))
  rbind(
    data.frame(n = n, est = "mean",   bias = mean(m) - mu, var = var(m)),
```

```
    data.frame(n = n, est = "median", bias = mean(md) - mu, var = var(md))
  )
}

res <- do.call(rbind, lapply(c(10, 50, 250, 1000), simulate_props))
res
```

We compare the sample mean and the sample median as estimators of the population mean μ (under a symmetric normal, so both target the same quantity). The mean is efficient; the median is less efficient but is also consistent.

Output & Results

```
     n    est        bias       var
1   10   mean  0.00285700 0.9066562
2   10 median  0.00162894 1.3963523
3   50   mean  0.00098014 0.1795648
4   50 median -0.00132211 0.2743110
5  250   mean -0.00083720 0.0359942
6  250 median -0.00038881 0.0557902
7 1000   mean  0.00006472 0.0090152
8 1000 median -0.00007128 0.0140841
```

Both estimators have bias near zero; variance shrinks at rate $1/n$ (the mean hits $\sigma^2/n = 9/n$; the median is higher by the factor $\pi/2 \approx 1.57$). The mean is thus more efficient by roughly 1.57.

Interpretation

For normally distributed data, the sample mean is both unbiased and efficient; choosing it over the median costs nothing and gains precision. For heavy-tailed data or data with outliers, the median is more robust, and the efficiency gap can reverse in the opposite direction. These are trade-offs that the three criteria make explicit.

Practical Tips

- Consistency is usually the bare minimum requirement; without it, more data does not help.
- Unbiasedness is a mathematical convenience that can be sacrificed for a smaller variance (lower MSE).
- The asymptotic efficiency of the MLE is why it is a default estimator in regular parametric models; the cost is specification error if the model is wrong.
- Efficiency is relative to a model: under misspecification, the "efficient" estimator may be inconsistent while a "less efficient" robust one remains consistent.
- Always compare estimators on the quantity that matters for the application – MSE, coverage of CIs, predictive error – not on a single property in isolation.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr